Smart Cities and Urban Technology

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Introduction to Smart Cities

A smart city is an urban area that uses digital technology, sensors, data analytics and connected infrastructure to improve the quality of life of its residents, make city services more efficient, reduce costs and protect the environment. In a smart city, devices such as traffic cameras, smart meters, air-quality sensors and water-level monitors collect real-time data, which is analysed at a central command centre to help officials make faster and better decisions. Smart cities are built on the Internet of Things (IoT), cloud computing, artificial intelligence, high-speed networks and open data. The idea does not mean only gadgets: it also includes good governance, citizen participation and sustainability.

Key Areas of a Smart City

Smart mobility improves transport through intelligent traffic signals, public transport tracking and electric vehicles. Smart energy uses smart meters, solar power and efficient street lighting. Smart water systems detect leaks and monitor quality. Smart waste management uses sensors in bins and tracked garbage trucks. Smart governance offers online services, grievance apps and open data. Public safety uses CCTV, emergency response systems and disaster alerts. Smart buildings and housing use energy-saving design. Healthcare and education services are also digitised.

Technologies in Smart Traffic Systems

Two technologies used in smart traffic systems are adaptive traffic signals and AI-enabled cameras. Adaptive traffic signal control uses sensors, such as inductive loops, radar or cameras, to count vehicles and adjust the length of green lights in real time, reducing waiting time and congestion. AI cameras with automatic number plate recognition (ANPR) detect traffic violations such as red-light jumping, speeding and not wearing helmets, and issue e-challans automatically. Other technologies include GPS tracking of buses, mobile apps with real-time traffic maps, electronic toll collection, variable message signs, smart parking sensors and Intelligent Transport Systems (ITS) integrated at a central control room. Advanced cities test vehicle-to-infrastructure (V2X) communication.

Global Examples

Singapore is often cited as a leading smart nation, with electronic road pricing, extensive data use and digital government services. Songdo in South Korea was built from scratch as a smart city. Barcelona uses smart lighting and sensors, and Amsterdam runs open-data and smart mobility projects. Dubai promotes smart government and drones.

India's Smart Cities Mission

India's Smart Cities Mission was launched by Prime Minister Narendra Modi on 25 June 2015. Its aim was to develop 100 cities across the country with core infrastructure, a clean and sustainable environment, and a decent quality of life through smart solutions. The Ministry of Housing and Urban Affairs implemented it. The central government committed about Rs. 48,000 crore over the mission period, with matching funds from states and city bodies, and cities were chosen through a competition in several rounds starting in January 2016. Each city created a Special Purpose Vehicle (SPV) to carry out projects, and nearly every city set up an Integrated Command and Control Centre (ICCC). Two approaches were used: area-based development, which retrofits or redevelops a selected area of the city, and pan-city solutions, which apply technology across the entire city. The mission period was extended until March 2025.

Smart Cities in Tamil Nadu and Chennai

Several Tamil Nadu cities, including Chennai, Coimbatore, Madurai, Tiruchirappalli, Thanjavur and Vellore, were selected under the mission. In Chennai, projects included the pedestrian plaza at Pondy Bazaar in T. Nagar, smart roads, solar rooftops, and an Integrated Command and Control Centre linked to thousands of CCTV cameras. Coimbatore and Madurai took up lake restoration, smart bus shelters and heritage area improvements. Surat, Indore, Pune and Bhopal are among the other Indian cities known for strong smart city projects, and Indore has repeatedly been ranked India's cleanest city.

Benefits

Smart cities can cut traffic jams, reduce energy and water waste, improve air quality, improve safety, deliver faster government services and give citizens real-time information through mobile apps. They also create jobs in technology and encourage innovation.

Challenges

Challenges include high cost and funding gaps, privacy concerns from surveillance and data collection, cybersecurity risks, the digital divide that can leave poorer residents behind, poor coordination between agencies and slow implementation. Critics argue that some projects focused on small areas rather than on the needs of the entire city.

Future

Future cities will rely on digital twins, which are virtual replicas of cities used for planning, AI-driven traffic management, electric and autonomous transport, renewable power and climate-resilient design. India's AMRUT and Urban Challenge Fund continue the work of improving urban infrastructure.

Quick Facts

- A smart city uses digital technology and data to improve services and quality of life.
- Adaptive traffic signals and AI-enabled cameras are two smart traffic technologies.
- India's Smart Cities Mission was launched on 25 June 2015 to develop 100 cities.
- Each smart city has an Integrated Command and Control Centre (ICCC).
- Chennai's Pondy Bazaar pedestrian plaza is a well-known smart city project.